Language in Visual Emotion Analysis: A Survey

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Abstract

Visual Emotion Analysis (VEA) aims to perceive the emotions of people towards an image, which plays a very important role in various areas of our daily life, especially in human-machine interaction, mental healthcare, education, and so on. To improve machines' ability to understand the emotion of an image, people have built a lot of methods/models. But recent studies have discovered that using only visual information has reached its limit. Therefore, a new modality is introduced to VEA —- language (text). In this paper, I will give a comprehensive overview of the field of VEA, and especially draw attention to the importance of text in VEA these days. We begin with a detailed introduction to VEA and how it relates to reality. We then review the basic sentiment theory, which is the foundation of today's VEA works. After that, we will list the VEA modeling in visual-only methods and discuss their limitations through the results from public sentiment image datasets. We also discuss how language plays an important role in VEA by introducing psychological theories first, and then we can see the improvement from using language in VEA through the results from datasets. Finally, we will discuss the limitation of text assistance, and we also introduce a framework for how to generate proper text to help VEA. Overall, this article endeavors to equip readers with a deeper understanding of the domain of VEA and to inspire further research and development in this captivating and rapidly evolving field.

1 Introduction

1.1 Visual Emotion Analysis

Visual Emotion Analysis (VEA) (often discussed under the broader umbrella of affective visual information analysis) studies how machines can infer, model, and reason about emotions from visual media—most commonly images and videos [WZW+23]. In this paper, we discuss emotions in images mainly. The long-term ambition of VEA is to endow AI systems with a form of "visual EQ," so that they can understand affective states and respond appropriately in human-facing contexts (e.g., interaction, assistance, safety, retrieval, entertainment).The topic of VEA is intrinsically interdisciplinary, pulling from computer vision, multimedia, psychology, data science, movement analysis, and the performing arts, precisely because emotion is not a purely visual construct and its interpretation depends on perception, context, and social convention.

Despite rapid progress in deep learning, VEA is in its infancy largely because emotion itself lacks a universally accepted definition and because emotional judgments are subjective and nuanced. We will further discuss in session 5.

1.2 Why Matters

Visual Emotion Analysis (VEA) matters because it adds an affective layer to computer vision: beyond recognizing objects and actions, systems can estimate how visual content expresses emotion (e.g., via faces/bodies) and evokes emotion in viewers. This capability is widely relevant because emotion is a core channel of human communication, and "visual emotional understanding" naturally connects to applications spanning art/design [LSAJ+12, LSN+16], mental health [FYL+20], ad [SGK+22], robotics and human–AI interaction [SW21, CSF+18, MPGP22], autonomous vehicles [CFZ22], and entertainment such as animation [YZL19] and gaming ~ [BCX24].

Concretely, VEA enables high-impact, user-facing decisions: in mental health care, deep CNN-based emotion analysis has been studied as a pathway to scalable mental-state screening and support [FYL+20]; in computational advertising, recognizing the emotions conveyed/elicited by ads can improve downstream tasks like ad understanding and better ad placement to optimize user experience and effectiveness [SGK+22]. In parallel, VEA can improve safety and interaction quality in settings like autonomous driving [CFZ22] (where subtle affective cues can signal risk or intent), and it can drive more adaptive experiences in interactive media—e.g., games ~ [BCX24] that adjust narrative or difficulty in response to inferred player frustration or sadness.

2 Basic sentiment theory

Basic Emotion theories. The discrete tradition frames emotion as a small set of qualitatively distinct states with characteristic functions and expressions. From an evolution–function perspective, Charles Darwin argued that emotional expressions are biologically grounded adaptatevant behavior [DP98]. From an embodied/physiological perspective, William James emphasizedto patterns of bodily arousal and action tendencies [Jam13]. Building on these assumptions, Paul Ekman proposed a neurocultural account in which several "basic" emotions are supported by a broadly shared affect program and are recognizable across cultures [EF78]. This line of thinking directly serves VEA tasks with discrete labels: it provides an interpretable target space for annotation, benchmarking, and error analysis, and it matches how many datasets are actually curated (i.e., categorical emotion tags). At the same time, the survey highlights well-known challenges—people can be inconsistent in discrete labeling and show overlap among reported emotions —so ood as an operational, task-driven approximation rather than a complete theory of affect.

Dimensional models. Dimensional accounts describe affect using continuous coordinates rather than categories. Albert Mehrabian and James A. Russell introduced the PAD state model (Pleasure–Arousal–Dominance) [MR74], which is closely related to the engineering-frienrousal–Dominance). Russell later emphasized core affect [Rus03], focusing on pleasantness (valence) and arousal as fundamental dimensions. PAD/VAD locate an emotion in a continuous space of valence/pleasure, arousal, and dominance/control.For VEA, this provides a lightweight way to describe emotion intensity and similarity and to interpret ambiguous cases.

Cognition-based theories. Cognitive views argue that emotions depend not only on physiology or stimulus cues but also on how events are interpreted. Magda Arnoldded the idea that cognition is necessary for emotion elicitation and that different individuals can experience different

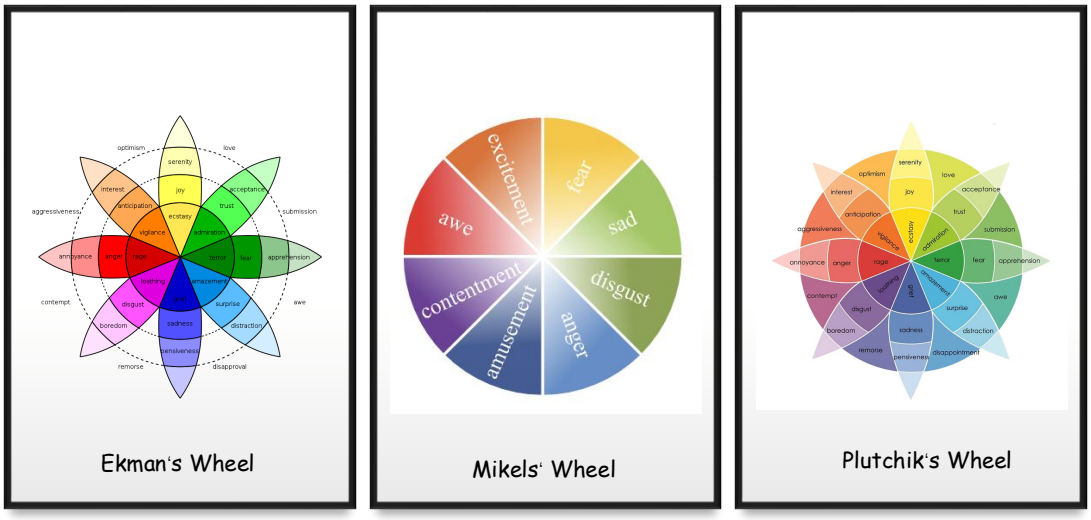

Figure 1: Ekman's, Mikels', and Plutchik's emotion wheels.

emotions in response to the same situation because they evaluate it differently [Arn60]. Extending this, Stanley Schachter' s two-factor theory links emotional experience to both arousal and its cognitive framing, showing that identical physiological arousal can be experienced as different emotions depending on context and interpretation [SS62]. For VEA, these theories motivate why contextual and semantic information (scene meaning, intent, social interaction, norms) is often indispensable: the same visual pattern can evoke different discrete labels once appraisal-relevant cues are considered. Practically, they justify context-aware architectures and evaluation setups (e.g., modeling scene–agent relations, uncertainty, and annotator variability) even when the output is still a discrete emotion class.

Emotion wheels. As shown in figure 1 There are 3 main wheel that used in VEA: Ekman' s, Plutchik' s and Mikels' wheels. Emotion wheels provide structured discrete affect taxonomies that are widely used in VEA. Plutchik' s wheel organizes a small set of primary emotions (and their mixtures/intensities) into a compact relational structure [Plu80]. Mikels' emotion wheel/taxonomy, developed in the context of emotion-eliciting visual stimuli, is frequently adopted in image-based emotion datasets to define an 8-class categorical space [MFL+05]. Ekman' s basic-emotion taxonomy (often operationalized as an Ekman-style 6-class set) remains a common choice for discrete emotion recognition because it yields a clear, interpretable label set tied to facial-expression universals [Ekm71].

3 Datasets

Before reviewing the methods, we first go through the dataset we usually use in VEA. As show in table 1. As Magda Arnoldded have said, due to subjectivity and the differences in

Table 1: Comparison between VEA datasets.

Dataset	**Image**	**Annotator (per image)**	**Model (category)**	**Image type**
IAPS	395	-	Mikels (8)	Natural
Abstract	280	14	Mikels (8)	Abstract
ArtPhoto	806	-	Mikels (8)	Artistic
Twitter I	1,269	5	Sentiment (2)	Social
Twitter II	603	3	Sentiment (2)	Social
Emotion6	1,980	**15**	Ekman (6)	Social
HECO	9,385	-	- (8)	Social
Flickr	60,745	3	Sentiment (2)	Social
Instagram	42,856	3	Sentiment (2)	Social
FI	23,308	5	Mikels (8)	Social
EMOTIC	18,316	3	Ekman (6)	Social
FlickrLDL	10,700	11	Mikels (8)	Social
TwitterLDL	10,045	8	Mikels (8)	Social
EmoSet	**118,102**	10	Mikels (8)	**Social, Artistic**

concepts among various cultures and ethnic groups, datasets in the VEA field generally adopt crowd sourcing annotation methods to label the emotions of images. Table 2 shows the quantity of each emotion category in 8 datasets.

IAPSa [LBC+99] contains 395 natural images and is organized under an 8-class emotion taxonomy based on the Mikels model. It is commonly used as a compact benchmark for affective understanding, while detailed per-image annotator statistics are not explicitly specified in the dataset summary.

Abstract [MH10] is a small-scale abstract image dataset with 280 samples, annotated under the Mikels 8-class emotion model. In terms of reliability, each image is labeled by 14 human annotators, providing relatively strong inter-subject agreement for abstract stimuli.

ArtPhoto [MH10] includes 806 artistic photos and follows the Mikels 8-class emotion categories. It targets emotion perception in artistic imagery, while the dataset summary does not provide a fixed number of annotators per image.

HECO [YHW+22] contains 9,385 social images and provides an 8-category emotion setting. The dataset summary does not report a fixed number of annotators per image, but its scale makes it suitable for training and evaluation under broader data regimes.

Flickr and Instagram.[KS16] Images in Flickr and Instagram [17] are crawled from the internet by searching emotional categories. After labeling by crowd-sourced human annotation, 60,745 and 42,856 affective images are preserved with sentiment labels (i.e., positive or negative).

Table 2: Comparison on image number per category, where purple indicates the maximum while green the minimum.

Dataset	Amusement	Anger	Awe	Contentment	Disgust	Excitement	Fear	Sadness	Total
IAPS	37 (9%)	8 (2%)	54 (14%)	63 (16%)	74 (19%)	55 (14%)	42 (11%)	62 (16%)	395
Artphoto	101 (13%)	77 (10%)	102 (13%)	70 (9%)	70 (9%)	105 (13%)	115 (14%)	166 (21%)	806
Abstract	25 (11%)	3 (1%)	15 (7%)	63 (28%)	18 (8%)	36 (16%)	36 (16%)	32 (14%)	228
FI	4942 (21%)	1266 (5%)	3151 (14%)	5374 (23%)	1658 (7%)	2963 (13%)	1032 (4%)	2922 (13%)	23308
EmoSet	19445 (16%)	10660 (9%)	15037 (13%)	16337 (14%)	10666 (9%)	19828 (17%)	13453 (11%)	12676 (11%)	**118102**

Specifically, each image is labeled by 3 workers, where the ground-truth is determined by a majority vote.

EMOTIC [KARL17] contains 18,316 social images and is aligned here with a 6-class Ekman-style emotion setting. For reliability, each image is annotated by 3 human annotators, supporting learning of emotion categories under contextual scenes.

FlickrLDL [YSS17] is a social-image dataset designed for learning emotion distributions, containing 10,700 images under the Mikels 8-class model. Each image is annotated by 11 annotators, which is particularly helpful for estimating richer label distributions beyond hard one-hot labels.

TwitterLDL [YSS17] contains 10,045 social images and is also formulated under the Mikels 8-class emotion model with label distribution learning. Each image is annotated by 8 annotators, balancing annotation cost and the quality needed to model soft emotion distributions.

EmoSet [YHD+23] is the largest dataset, which contains 118,102 images and maintains a balanced distribution across 8-class emotions based on the Mikels model. In terms of reliability, each image is annotated by 10 human annotators.

FI [YLJY16] is collected from Flickr and Instagram by searching eight emotion categories from the Mikels model as keywords. It contains 22,683 images, and their eight emotion categories can be split into positive (amusement, awe, contentment, excitement) and negative (anger, disgust, fear, sadness). The label of each image is determined by the consensus of at least three out of Amazon Mechanical Turk (AMT) workers.

EmotionROI [PSGC16] is retrieved from Flickr by querying six Ekman emotion words and their synonyms (i.e., anger, disgust, fear, joy, sad, surprise). Besides, joy and surprise are positive labels, and anger, disgust, fear, sad are negative labels. The label distribution of EmotionROI is class-balanced (330 images per emotion), resulting in 1,980 images in total.

Twitter I [BJC+13b] and **Twitter II** [YLJY15] are two small visual emotional benchmarks collected from the Twitter website. Both are labeled with positive and negative. Twitter I contains 1,269 images with 5 AMT annotators per image, while Twitter II contains 603 images with 3 AMT annotators per image.

Because **Emoset, FI, EmotionROI, Twitter I and Twitter II** are widely used benchmark datasets, we mainly report our model' s performance on it in the following experiments.

Table 3: Quantitative comparison of MCRStimuli and competing methods on different visual emotion datasets. The best results are denoted in **bold** and the second-best results are underlined. The reported values for the competing methods are taken directly from the original publication.

Category	Method	EmoSet		FI		EmotionROI		Twitter I	Twitter II
Number of classes		8	2	8	2	6	2	2	2
Visual-only	AlexNet [KSH12a]	0.6780	0.8928	0.5813	0.6863	0.4141	0.7160	0.7324	0.7566
	ResNet101 [HZRS16a]	0.7487	0.9384	0.6616	0.7576	0.5160	0.7392	0.7813	0.7823
	WSCNet [SYC+19]	0.7632	0.9416	0.7007	–	0.5825	–	0.8425	0.8135
	SOLVER [YGL+21]	–	–	0.7233	–	0.6212	–	0.8543	0.8319
Language-assist	MERL [ZKL+16]	–	–	0.7650	0.9116	–	0.8560	–	–
	SentiBank [BJC+13a]	–	–	0.4923	0.5647	0.3524	0.6618	0.6663	0.6593
	DeepSentibank [CBDC14]	–	–	0.5154	0.6154	0.4253	0.7011	0.7125	0.7023
	MRCNN [RLZX19b]	0.7539	0.9228	0.7546	0.8751	–	0.8294	–	–
	OBGCN [WZD+21]	–	–	–	0.8871	–	0.8429	0.8965	0.8268
	StimuliVEA [YLW+21]	0.7840	0.9458	0.7242	–	0.6162	–	–	–
	MDAN [XWWL22]	0.7575	0.9371	0.7641	0.9108	0.6166	0.8462	–	–
VLM-assisted	CLIP [RKH+21]	0.6976	0.9074	0.7784	0.8906	0.6565	0.8773	0.8710	0.8145
	CoCoOp [ZYLL22]	0.8031	0.9333	0.7746	0.9124	0.7109	0.8910	0.9163	0.8294
	SimEmotion [DWS+23]	0.7906	0.9428	0.8033	0.9542	0.7054	0.9040	0.8976	0.8421
	PT-DPC [DWS+24]	0.7713	0.9246	0.7807	0.9389	0.6970	0.8855	0.9094	0.8250
	MASANet [CQO+24]	–	–	0.7916	0.9507	0.7323	0.9175	0.9216	0.8870
	MVP [XWY+24]	0.8192	**0.9644**	**0.8276**	**0.9698**	0.7189	**0.9255**	0.9203	0.8821
	EmoViT [XPT+24]	**0.8336**	–	0.6809	–	0.5387	–	–	–
	EmoVLM-KD [LLP25]	0.7983	–	0.7951	–	–	–	–	–
	ESCNet [ZZGZ25]	–	–	0.8086	0.9571	**0.7399**	0.9186	**0.9246**	**0.8926**

Note: Best results are in **bold**; second-best results are underlined.

4 Methods in VEA

A schematic overview of the three paradigms is provided in fig 2. We categorize existing methods in VEA into three paradigms according to the source of semantic evidence and the modeling architecture. **(1) Visual-only** approaches rely solely on a visual encoder backbone and extract emotion-relevant cues only from the image itself. Even when equipped with components such as object detection or region reasoning, these methods remain bounded by visual evidence without incorporating any external information; consequently, their performance often saturates due to the intrinsic ambiguity and subjectivity of affective perception. **(2) Language-assisted approaches** still adopt a single visual encoder as the main backbone, but inject external priors into visual representation learning, such as sentiment knowledge from language resources, human affective rules, or psychological theories of emotion. These priors guide the model toward more semantically grounded affect inference while keeping the overall architecture unimodal at inference time. **(3) VLM-assisted approaches** explicitly introduce text as an additional modality. Textual inputs can be formulated as natural-language hard prompts or learnable soft prompts, and

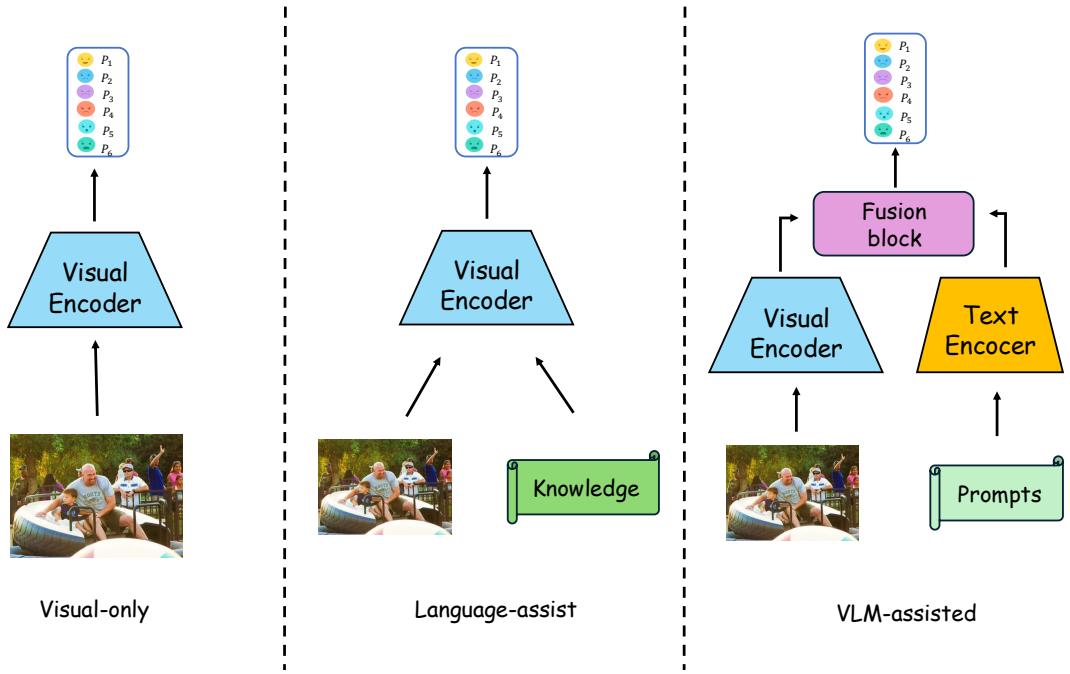

Figure 2: The overlook framework of 3 different methods in VEA

are processed by a text encoder. A fusion module then integrates visual and textual representations, substantially enriching the available semantic signals and expanding the feature dimensions that the model can exploit. This paradigm is particularly suitable for emotion analysis, where reliable prediction often requires complementary cues beyond raw visual appearance.

4.1 Visual-only methods

Early handcrafted and interpretable representations. Early work in visual emotion analysis (VEA) largely followed a classic "feature + classifier" pipeline, extracting handcrafted descriptors from images or faces (e.g., HOG/LBP-style texture and gradient cues) and training conventional classifiers such as SVM for affect prediction [W+23]. In parallel, many studies emphasized explainable visual representations inspired by psychology and art theory, modeling affective cues through geometry and composition (e.g., symmetry, depth, figure–ground, scene complexity) as well as shape attributes (e.g., roundness vs. angularity) to rationalize why an image evokes certain emotions [ZGJ+14, W+23, MH10].

Deep visual backbones for end-to-end learning. With the rise of deep learning, VEA increasingly adopted end-to-end representation learning using standard visual backbones. CNN models such as AlexNet and deeper architectures like ResNet became common choices for extracting affect-discriminative features directly from pixels [KSH12b, HZRS16b, W+23]. For sequential emotion understanding (e.g., videos), RNN/LSTM-style models were used to capture temporal dynamics over frame-level visual features [W+23]. More recently, Transformer-based vision encoders (e.g., ViT) have also been widely used as generic visual backbones for VEA due to their strong global representation capacity [D+21].

Task-specific visual modeling without external knowledge. Recent visual-only meth-

ods go beyond generic backbones by introducing task-specific visual modeling, such as region localization, object/scene interaction reasoning, and distribution learning, while still relying solely on image-derived signals at inference. **WSCNet** couples visual sentiment classification with weakly supervised detection/localization of sentiment-relevant regions, effectively learning "where to look" using only visual supervision signals [SYC+19]. **SOLVER** explicitly models scene–object interrelations by reasoning over detected visual entities and refining emotion-enhanced object features, but its evidence remains fully grounded in the input image [YGL+21]. **Hierarchical CNN** variants further fuse multi-level representations across layers/regions to capture affective cues at different granularities (global aesthetics vs. local stimuli), providing a structured yet unimodal alternative to external-knowledge injection [RLZX19a].

4.2 The role of language in VEA

In VEA, Intra-class appearance is highly variable, while inter-class visual cues substantially overlap, so models trained on images alone often collapse toward coarse affective dimensions such as valence rather than learning robust category boundaries. Language mitigates this issue by injecting explicit semantic constraints and concept-level structure, shifting learning from fitting visual correlations to separating categories according to their definitional meaning. [LMS15] For example, facial patterns such as a wrinkled nose and tightened mouth, a furrowed brow with compressed lips, widened eyes with tension, or lowered gaze with tightened facial muscles can be visually compatible with multiple negative categories, including disgust, anger, fear, and sadness, especially under occlusion, low resolution, or domain shift. Providing text supervision in the form of emotion-word sets, category definitions, synonym expansions, or generated category rationales makes the discriminative semantics explicit, reducing confusions among visually similar classes and supporting cross-dataset label alignment when taxonomies differ.

A second fundamental limitation in VEA is that emotion inference depends on contextual meaning assignment. Emotions are not determined solely by visible cues, but emerge from integrating sensory information with situation-dependent interpretation supported by learned emotion concepts. Language contributes to this process by supplying event context, causal structure, intent, and relational variables that are frequently absent from the image. [LMS15] For example, the same visual evidence of tears, lowered head, and muscular tension can reasonably map to sadness, relief, gratitude, or joy, depending on the triggering event. If accompanying text indicates bereavement, loss of work, or a severe setback, the interpretation tends to converge toward sadness; if it indicates reunion after long separation or achievement after prolonged effort, the interpretation may instead converge toward relief or joy. In practice, incorporating titles, comments, OCR text, captions, or language-model-generated event and cause descriptions can expose latent contextual factors, transforming visual features into evidence that is interpretable under a specific situation model. This mechanism is a primary driver of the gains observed in VLM-assisted VEA, particularly on samples that admit multiple plausible label assignments from vision alone.

4.3 Language-assist methods

Language-based sentiment information. A representative line of work injects language-driven sentiment cues into visual representation learning, aiming to bridge the affective gap with richer semantic support. **MERL** leverages sentiment-related linguistic information as auxiliary guidance, so that the learned visual embedding space aligns better with emotion-relevant semantics for this inherently abstract task [ZKL+16]. **SentiBank** is a milestone approach that introduces *adjective–noun pairs* (ANPs) (e.g., "crying baby") as sentiment-biased visual concepts and provides a large-scale sentiment ontology with detectors for thousands of ANPs, offering a high-level semantic interface that is both emotion-informative and visually detectable [BJC+13a]. **DeepSentiBank** further upgrades this idea by training deep CNN-based concept classifiers for sentiment-related visual concepts, improving the capacity and robustness of ANP-style sentiment concept recognition under the deep learning paradigm [CBDC14].

External knowledge injection and rule-inspired modeling. Beyond language-derived sentiment concepts, some methods explicitly incorporate external knowledge or structural priors to enrich purely visual representations. **MRCNN** adopts a multi-level region-based framework and enhances discriminative power by incorporating external knowledge, motivated by the observation that emotion is abstract and difficult to capture from raw images alone [RLZX19b]. **OBGCN** introduces a rule-inspired perspective: it models the *interactive characteristics of objects in the sentimental space* under human sentimental principles (i.e., different regions/objects contribute differently to sentiment) and employs a graph-based network to capture object–graph relationships within an image for more faithful affect reasoning [WZD+21].

Psychological and cognitive knowledge. Another stream draws from psychology and cognition to construct emotion-aware inductive biases. **Stimuli-aware** is motivated by psychological views of emotion elicitation and explicitly emphasizes emotional *stimuli* (e.g., color/object/face cues) and their interactions, introducing a stimuli-aware pipeline that encourages the model to attend to psychologically meaningful evidence rather than treating the whole image uniformly [YLW+21]. **MDAN** further formalizes psychological structure by leveraging emotion hierarchy and correlations across affective/semantic levels, introducing an emotion category model grounded in psychological knowledge to guide multi-level prediction and reduce the affective gap [XWWL22].

4.4 VLM-assisted Methods

Overview and motivation. A growing body of work leverages vision–language models (VLMs) to alleviate the intrinsic limitation of single-stream visual encoders in VEA. After years of architectural refinement, purely visual pipelines have approached a performance ceiling, largely because affect inference is abstract and often under-determined from pixels alone. VLM-assisted methods introduce an additional text encoder and explicitly incorporate textual evidence as a new modality, substantially expanding the semantic bandwidth available for prediction and enabling notable performance gains. In general, these approaches improve VEA through three key design choices: (i) how to formulate effective prompts, (ii) how to select or adapt a compatible text encoder with the visual backbone, and (iii) how to fuse the two modalities. In what follows,

we focus on prompt design, which has become a dominant factor in recent progress.

Soft prompt learning. Soft prompt methods optimize learnable context vectors (rather than discrete words) to steer a frozen or lightly-tuned VLM toward downstream emotion categories. CoOp learns a set of continuous context tokens that are concatenated with class names to form class-specific prompts, enabling efficient adaptation with minimal parameters [ZYLL21]. CoCoOp further improves generalization by conditioning the prompt on the input image, producing instance-adaptive prompts that mitigate overfitting to the training distribution [ZYLL22]. PT-DPC extends soft prompting with diversified and learnable prompt components, aiming to cover richer affective variations via prompt composition and regularized optimization [DWS+24]. Despite their efficiency, soft prompts often exhibit unstable training dynamics: performance can fluctuate noticeably across runs and is sensitive to the initialization/anchor point of the prompt vectors. Moreover, their continuous prompts are difficult to interpret and do not naturally provide explicit reasoning, which limits their transparency for affect understanding.

Hard prompts with label-centric text. Another line of work employs natural-language (hard) prompts, typically built around emotion labels or short label descriptions, to query a pretrained VLM. CLIP provides the canonical baseline by matching image features with text features derived from label prompts, enabling zero-shot or prompt-tuned recognition [RKH+21]. SimEmotion incorporates emotion-related entity knowledge and designs prompt templates to better align label semantics with emotion-indicative visual evidence, improving robustness over naive label prompting [DWS+23]. MASANet further enhances label-based prompting by introducing multi-aspect semantic alignment, aiming to align different semantic facets of emotion labels with corresponding visual cues [CQO+24]. However, prompts that rely primarily on class labels (or minimal label text) can induce a form of semantic rigidity: the model may become overly biased toward surface-level label semantics and may fail to internalize the latent reasoning process that connects visual stimuli to affective states, leaving room for further improvement.

Hard prompts with LLM-generated reasoning text. Recent methods therefore move beyond label-centric prompts and use large language models (LLMs) to generate richer affective descriptions, rationales, or multi-sentence reasoning text as the textual modality. EmoViT leverages VLM-style prompting while emphasizing richer emotion-related textual context, strengthening the model's ability to capture nuanced affective semantics beyond short labels [XPT+24]. EmoVLM-KD distills knowledge from VLM/LLM signals and employs knowledge distillation to transfer reasoning-enhanced supervision into the student model, improving affect understanding under limited annotations [LLP25]. ESCNet further exploits structured emotion semantic cues and richer textual explanations to better couple emotion reasoning with visual evidence, leading to stronger generalization and improved accuracy [ZZGZ25]. By providing explicit, reasoning-oriented text, these approaches encourage the model to learn the internal affective logic—i.e., how visual stimuli, context, and semantics jointly imply a target emotion—which is especially beneficial for VEA. As evidenced by the quantitative gains in Table 3, reasoning-enriched hard prompts can substantially strengthen model performance across datasets, suggesting that improving textual semantics and interpretability is a key driver for the next stage of progress.

Mixed prompting: combining hard and soft prompts. MVP adopts a hybrid prompting

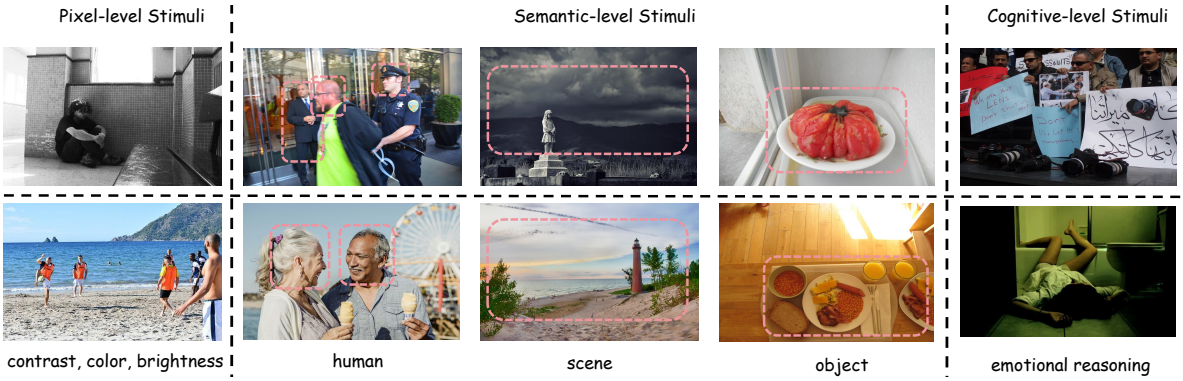

Figure 3: Illustration of visual examples which are associated with different levels of stimuli. (see text for details)

strategy that integrates discrete (hard) prompt components with learnable (soft) prompt tokens, aiming to exploit the complementary strengths of both paradigms [XWY+24]. Concretely, MVP introduces multiple "views" of emotional prompts, where textual inputs provide explicit semantic anchors while optimized prompt parameters adapt the prompt space to dataset-specific affect distributions, and the final prediction benefits from multi-view aggregation. This mixed design is often more flexible than purely label-centric hard prompts, as it can inject task-relevant semantic priors while retaining the adaptability and parameter efficiency of soft prompt tuning. However, MVP also inherits practical limitations from soft prompting: training can still be sensitive to initialization and optimization settings, and the learned prompt vectors remain less interpretable than fully natural-language reasoning prompts. Moreover, because its textual side is still largely constrained by predefined templates and prompt parameterization, it may not capture explicit, human-readable affect reasoning as directly as approaches that leverage LLM-generated long-form rationales.

5 Limitation and Discussion

A single image may not admit a unique ground-truth label. Conventional supervised learning typically assumes that each sample is associated with one certain label. However, in VEA, emotional perception is inherently influenced by cognition and subjective appraisal, implying that individuals with different cultural backgrounds and personal experiences may legitimately report different emotions for the same image. Moreover, several emotion categories exhibit substantial semantic overlap (e.g., amusement versus awe), further increasing annotation variability. As a result, it is often unrealistic to expect an absolutely correct and unique label for every sample. This highlights the need for models to move beyond surface-level pattern matching and instead develop a principled understanding of how visual content elicits affect. Learning robust affective reasoning that links image content to emotion is therefore an important direction for future research.

Label supervision alone remains insufficient for learning affective reasoning. A categorical label provides only a single token of supervision and conveys limited information

about the underlying causes of an emotional response. Even when approaches such as SimEmotion expand a single emotion label into a small set of related emotion words using an emotion taxonomy (e.g., Plutchik's wheel), the supervision signal remains weak in terms of explaining why the image evokes the target emotion. Consequently, models may learn to reproduce emotion words without capturing the visual evidence and causal structure that support the prediction. Recent methods such as EmoViT and ESCNet leverage large vision–language models to generate reasoning-oriented prompts, yet these prompts are often still strongly conditioned on the ground-truth label, which can encourage shortcut learning and memorization rather than genuine affective understanding. In addition, emotion can be elicited by multiple factors within an image, including humans, scenes, and objects, as well as their attributes and interactions, leading to diverse and compositional affective cues. This motivates the development of a general and comprehensive framework for identifying and reasoning about sentiment stimuli in a structured manner.

To address this challenge, we propose the Context-aware Reasoning Stimuli framework (MCRStimuli), which aims to comprehensively decode image-inherent sentiment stimuli at multiple levels, including pixel-level, semantic-level, and cognitive-level cues. These three tiers reflect successive stages of visual abstraction and affective elaboration, corresponding to different contexts, as illustrated in Figure **??**. Unlike prior methods that primarily rely on pre-trained vision models to extract emotional cues, we introduce a self-generation pipeline that leverages a large vision–language model to infer stimuli at different levels. Specifically, given an input image, we first prompt the model to describe colorfulness, brightness, and contrast as pixel-level stimuli. We then identify the dominant scene, salient objects, and human subjects, and characterize their interactions to form semantic-level stimuli. Finally, these objective stimuli are integrated to produce linguistically grounded affective reasoning, which constitutes the cognitive-level stimuli.

Balancing visual and textual representations remains a key challenge. In multimodal learning, effective fusion of visual and textual information is critical. In practice, the two modalities may differ substantially in representation richness and optimization dynamics, which can cause the model to over-rely on the more informative modality and underutilize the weaker one, thereby limiting cross-modal complementarity. Achieving stable and balanced learning requires careful design across three components: (a) the visual and text encoders, (b) the construction and quality of textual inputs (e.g., prompts, captions, or rationales), and (c) the fusion module that integrates the two modalities. Each component plays a decisive role in determining whether the model can jointly exploit both modalities and learn robust multimodal affective representations.

6 Conclusion

This survey provides an overview of research on Visual Emotion Analysis, with an emphasis on image-based settings. We first review foundational affective theories that motivate and inform VEA label design and modeling assumptions. We then summarize representative modeling paradigms, ranging from visual-only approaches to vision–language-model-assisted methods, and

highlight the central role of language in improving affective understanding. In particular, we analyze prompting-based strategies that leverage text to support emotion recognition and discuss their benefits and limitations. Finally, we outline open challenges and promising future directions, including label ambiguity, the insufficiency of label-only supervision for affective reasoning, and the need for principled multimodal fusion.

Our goal is to equip researchers and practitioners with essential background and practical insights for addressing the remaining open problems in this field. We encourage interested readers to consult the cited literature for deeper technical details and broader perspectives. We also recommend collaboration across complementary areas of expertise, as progress in VEA often requires integrating advances in vision, language modeling, affective computing, and human-centered evaluation.

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